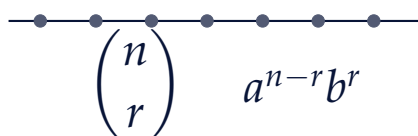


$$(a + b)^n$$



A horizontal line with seven dots. Below the third dot from the left is the binomial coefficient $\binom{n}{r}$. Below the fourth dot from the left is the term $a^{n-r}b^r$.

Binomial Theorem Practice Question Set

Positive Integer Powers & the Extended (General) Binomial Series

23 Questions • Fully Worked Solutions

Instructions: A key formulas and method-notes page follows this cover page — read it carefully before starting, especially the rule for combining two or more expansions in the same question. Questions are ordered from Easy to Medium to Hard. Attempt each question first; the full worked solution is on the page immediately following, boxed in teal.

Key Formulas & Method Notes

1. The Binomial Theorem (positive integer n)

For $n \in \mathbb{Z}^+$,

$$(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r = \binom{n}{0} a^n + \binom{n}{1} a^{n-1} b + \cdots + \binom{n}{n} b^n,$$

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

This is a *finite* sum of exactly $n + 1$ terms, valid for **all** values of a and b — there is no restriction on x to worry about here.

2. The Extended (General) Binomial Series — negative or fractional n

When the index n is **negative** or a **fraction** (not a positive integer), the theorem no longer terminates. Instead, for the bracket $(1 + x)^n$ with *any* $n \in \mathbb{Q}$,

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$$

This is an **infinite** series, and (unlike Formula 1) it is only valid — i.e. only converges to the correct value — when $|x| < 1$.

3. Crucial technique: always make the leading term 1

The series in Formula 2 only applies to a bracket that already starts with a bare 1, i.e. $(1 + \cdots)^n$. If instead you are asked to expand $(a + bx)^n$ with n negative or fractional, you **must first factor a out** of the bracket so that what remains inside starts with 1:

$$(a + bx)^n = a^n \left(1 + \frac{b}{a}x\right)^n = a^n \left[1 + n \left(\frac{b}{a}x\right) + \frac{n(n-1)}{2!} \left(\frac{b}{a}x\right)^2 + \cdots\right].$$

Never apply the series formula directly to $(a + bx)^n$ with $a \neq 1$ — always normalize first. (This step is essential in Questions 19–23 of this set.)

4. Validity (the restriction on x)

After normalizing as in Formula 3, the series for $\left(1 + \frac{b}{a}x\right)^n$ converges only when

$$\left|\frac{b}{a}x\right| < 1 \iff |x| < \left|\frac{a}{b}\right|.$$

This restriction should always be stated as part of a complete answer whenever the extended series (Formula 2) is used — it is **not** needed for Formula 1 (positive integer n), since that expansion is always exact and finite.

5. Combining two or more expansions: the “most restrictive wins” rule

A question will sometimes ask you to combine (multiply, divide, add or subtract) **two or more** extended binomial expansions in the same problem, each with its *own* individual validity condition on x . The combined expression is only valid where **every one** of the individual expansions is valid simultaneously — that is, on the *intersection* of all the individual ranges. Since each condition has the form $|x| < (\text{some bound})$, the intersection is simply the **smallest** (most restrictive) of the individual bounds.

Worked illustration (see Question 21): combining $(1 - 2x)^{-1/2}$, valid for $|x| < \frac{1}{2}$, with $(1 - 4x)^{-1/2}$, valid for $|x| < \frac{1}{4}$, gives a product that is valid only for

$$|x| < \min\left(\frac{1}{2}, \frac{1}{4}\right) = \frac{1}{4},$$

not $\frac{1}{2}$ — the smaller (more restrictive) bound always sets the limit for the combined expression. As a rule of thumb:

- Write down the individual restriction for *each* bracket in the expression separately.
- The overall restriction is whichever of these is the **smallest** range (i.e. the largest lower bound on how small $|x|$ must be) — never the largest or an average of the two.

Questions 19, 21, 22 and 23 in this set all involve normalizing, expanding and (in several cases) combining more than one expansion, so they are excellent practice for this rule.

Section 1

The Binomial Theorem

Positive Integer Powers

Questions 1–18

Every expansion in this section uses Formula 1 from the notes page: n is always a positive integer, the expansion is a *finite* sum, and it holds for **all** values of x — no validity restriction is ever needed.

Question 1

[Maximum mark: 4]

No Calculator

Difficulty: Easy

Find the coefficient of x^8 in the expansion of $\left(x^2 - \frac{2}{x}\right)^7$.

Question 1

Solution

The general term in the expansion of $\left(x^2 - \frac{2}{x}\right)^7$ is

$$T_{r+1} = \binom{7}{r} (x^2)^{7-r} \left(-\frac{2}{x}\right)^r = \binom{7}{r} (-2)^r x^{14-2r-r} = \binom{7}{r} (-2)^r x^{14-3r}.$$

For the term in x^8 , set $14 - 3r = 8$, so $r = 2$. The coefficient is

$$\binom{7}{2} (-2)^2 = 21 \times 4 = \boxed{84}.$$

Question 2

[Maximum mark: 4]

No Calculator

Difficulty: Easy

Determine the first three terms of $(1 - 2x)^{11}$ in ascending powers of x , giving each term in its simplest form.

Question 2

Solution

Since 11 is a positive integer, the ordinary binomial theorem applies:

$$(1 - 2x)^{11} = \sum_{r=0}^{11} \binom{11}{r} (1)^{11-r} (-2x)^r = \sum_{r=0}^{11} \binom{11}{r} (-2)^r x^r.$$

The first three terms (constant, x^1 , x^2) are

$$r = 0 : \binom{11}{0} (-2)^0 = 1, \quad r = 1 : \binom{11}{1} (-2)^1 = 11 \times (-2) = -22,$$

$$r = 2 : \binom{11}{2} (-2)^2 = 55 \times 4 = 220.$$

So

$$(1 - 2x)^{11} = \boxed{1 - 22x + 220x^2 - \dots}$$

Question 3

[Maximum mark: 4]

Calculator

Difficulty: Easy

Find the coefficient of x^6 in the expansion of $(2x - 5)^9$.

Question 3

Solution

The general term is

$$T_{r+1} = \binom{9}{r} (2x)^r (-5)^{9-r}.$$

For the term in x^6 , take $r = 6$:

$$\binom{9}{6} (2)^6 (-5)^3 = 84 \times 64 \times (-125).$$

Computing, $84 \times 64 = 5376$, and $5376 \times (-125) = \boxed{-672\,000}$.

Question 4

[Maximum mark: 5]

No Calculator

Difficulty: Easy

Find the term independent of x in the binomial expansion of $\left(2x^2 + \frac{1}{2x^3}\right)^{10}$.

Question 4

Solution

The general term is

$$\begin{aligned} T_{r+1} &= \binom{10}{r} (2x^2)^{10-r} \left(\frac{1}{2x^3}\right)^r = \binom{10}{r} 2^{10-r} \left(\frac{1}{2}\right)^r x^{2(10-r)-3r} \\ &= \binom{10}{r} 2^{10-2r} x^{20-5r}. \end{aligned}$$

The term is independent of x when $20 - 5r = 0$, i.e. $r = 4$. The term is

$$\binom{10}{4} 2^{10-8} = 210 \times 4 = \boxed{840}.$$

Question 5

[Maximum mark: 5]

No Calculator

Difficulty: Easy

Find the value of the constant term in the expansion of $x^4 \left(x + \frac{3}{x^2} \right)^5$.

Question 5

Solution

The general term of $\left(x + \frac{3}{x^2}\right)^5$ is

$$\binom{5}{r} x^{5-r} \left(\frac{3}{x^2}\right)^r = \binom{5}{r} 3^r x^{5-3r}.$$

Multiplying by the factor x^4 outside the bracket, the overall power of x is $4 + 5 - 3r = 9 - 3r$. This is zero when $r = 3$, giving the constant term

$$\binom{5}{3} 3^3 = 10 \times 27 = \boxed{270}.$$

Question 6

[Maximum mark: 7]

Calculator

Difficulty: Easy

Bob invests 1000 dinar in an account which pays a nominal annual interest rate of 4% compounded quarterly. The amount of money in the account after one complete year can be written as $1000(1 + k)^4$ where $k \in \mathbb{Q}$.

- (a) Write down the value of k . [1]
- (b) Expand and simplify $(1 + x)^4$. [2]
- (c) Hence or otherwise, find the amount of money in the account after one complete year, giving your answer correct to the nearest dinar. [4]

Question 6

Solution

(a) A nominal annual rate of 4% compounded quarterly gives a quarterly rate of $4\% \div 4 = 1\%$, so

$$k = \boxed{0.01}.$$

(b) By the binomial theorem,

$$\begin{aligned}(1+x)^4 &= \binom{4}{0} + \binom{4}{1}x + \binom{4}{2}x^2 + \binom{4}{3}x^3 + \binom{4}{4}x^4 \\ &= \boxed{1 + 4x + 6x^2 + 4x^3 + x^4}.\end{aligned}$$

(c) Substituting $x = k = 0.01$ into the expansion from (b):

$$(1.01)^4 = 1 + 4(0.01) + 6(0.01)^2 + 4(0.01)^3 + (0.01)^4 = 1.04060401.$$

So the amount after one year is

$$1000 \times 1.04060401 = 1040.60401 \approx \boxed{1041 \text{ dinar}}.$$

Question 7

[Maximum mark: 5]

No Calculator

Difficulty: Medium

In the expansion of $(x + k)^7$, where $k \in \mathbb{R}$, the coefficient of the term in x^5 is 63. Find the possible values of k .

Question 7

Solution

The general term is $T_{r+1} = \binom{7}{r} x^{7-r} k^r$. The term in x^5 corresponds to $7 - r = 5$, i.e. $r = 2$, giving coefficient

$$\binom{7}{2} k^2 = 21k^2.$$

Setting this equal to 63:

$$21k^2 = 63 \implies k^2 = 3 \implies \boxed{k = \pm\sqrt{3}}.$$

Question 8

[Maximum mark: 5]

No Calculator

Difficulty: Medium

Consider the expansion of $\left(8x^3 - \frac{1}{2x}\right)^n$ where $n \in \mathbb{Z}^+$. Determine all possible values of n for which the expansion has a non-zero constant term.

Question 8

Solution

The general term is

$$\begin{aligned} T_{r+1} &= \binom{n}{r} (8x^3)^{n-r} \left(-\frac{1}{2x}\right)^r = \binom{n}{r} 8^{n-r} \left(-\frac{1}{2}\right)^r x^{3(n-r)-r} \\ &= \binom{n}{r} 8^{n-r} \left(-\frac{1}{2}\right)^r x^{3n-4r}. \end{aligned}$$

A constant term occurs when $3n - 4r = 0$ for some integer $0 \leq r \leq n$, i.e. $r = \frac{3n}{4}$. Since the binomial coefficient $\binom{n}{r}$ and the numerical factor $8^{n-r} \left(-\frac{1}{2}\right)^r$ are always non-zero, the constant term exists (and is automatically non-zero) exactly when $r = \frac{3n}{4}$ is an integer, i.e. when n is a multiple of 4:

$$\boxed{n = 4k, \quad k \in \mathbb{Z}^+} \quad (n = 4, 8, 12, 16, \dots).$$

Question 9

[Maximum mark: 5]

No Calculator

Difficulty: Medium

Consider the expansion of $(3 + x^2)^{n+1}$, where $n \in \mathbb{Z}^+$. Given that the coefficient of x^4 is 20412, find the value of n .

Question 9

Solution

The general term is

$$T_{r+1} = \binom{n+1}{r} 3^{n+1-r} (x^2)^r = \binom{n+1}{r} 3^{n+1-r} x^{2r}.$$

The term in x^4 corresponds to $2r = 4$, i.e. $r = 2$, giving coefficient

$$\binom{n+1}{2} 3^{n-1}.$$

Setting this equal to 20412 and solving by trial (since n is a positive integer):

$$\binom{n+1}{2} 3^{n-1} = 20412.$$

Trying $n = 7$: $\binom{8}{2} 3^6 = 28 \times 729 = 20412$. ✓

$$\boxed{n = 7}.$$

Question 10

[Maximum mark: 6]

No Calculator

Difficulty: Medium

The coefficient of x^2 in the expansion of $\left(\frac{1}{x} + 5x\right)^8$ is equal to the coefficient of x^4 in the expansion of $(a + 5x)^7$, $a \in \mathbb{R}$. Find the value of a .

Question 10

Solution

Left side. The general term of $\left(\frac{1}{x} + 5x\right)^8$ is

$$\binom{8}{r} \left(\frac{1}{x}\right)^{8-r} (5x)^r = \binom{8}{r} 5^r x^{2r-8}.$$

The term in x^2 needs $2r - 8 = 2$, i.e. $r = 5$, giving coefficient $\binom{8}{5} 5^5 = 56 \times 3125 = 175000$.

Right side. The general term of $(a + 5x)^7$ is $\binom{7}{s} a^{7-s} (5x)^s$. The term in x^4 needs $s = 4$, giving coefficient $\binom{7}{4} a^3 5^4 = 35 \times 625 a^3 = 21875 a^3$.

Equate:

$$175000 = 21875 a^3 \implies a^3 = 8 \implies \boxed{a = 2}.$$

Question 11

[Maximum mark: 6]

No Calculator

Difficulty: Medium

Consider the expansion of $\frac{(ax + 1)^9}{21x^2}$, where $a \neq 0$. The coefficient of the term in x^4 is $\frac{8}{7}a^5$. Find the value of a .

Question 11

Solution

The general term of $(ax + 1)^9$ is $\binom{9}{r}(ax)^r = \binom{9}{r}a^r x^r$. Dividing the whole expansion by $21x^2$ shifts every power of x down by 2, so the term contributing to x^4 in $\frac{(ax + 1)^9}{21x^2}$ comes from $r = 6$ in $(ax + 1)^9$:

$$\frac{1}{21} \binom{9}{6} a^6 x^{6-2} = \frac{84}{21} a^6 x^4 = 4a^6 x^4.$$

Setting the coefficient equal to $\frac{8}{7}a^5$:

$$4a^6 = \frac{8}{7}a^5.$$

Since $a \neq 0$, divide both sides by a^5 :

$$4a = \frac{8}{7} \implies \boxed{a = \frac{2}{7}}.$$

Question 12

[Maximum mark: 6]

No Calculator

Difficulty: Medium

Consider the expansion of $(2 + x)^n$, where $n \geq 3$ and $n \in \mathbb{Z}$. The coefficient of x^3 is four times the coefficient of x^2 . Find the value of n .

Question 12

Solution

The coefficient of x^k in $(2+x)^n$ is $\binom{n}{k}2^{n-k}$. The condition states

$$\binom{n}{3}2^{n-3} = 4\binom{n}{2}2^{n-2}.$$

Divide both sides by 2^{n-3} :

$$\binom{n}{3} = 4\binom{n}{2} \times 2 = 8\binom{n}{2}.$$

Using $\binom{n}{3} = \binom{n}{2} \cdot \frac{n-2}{3}$:

$$\binom{n}{2} \cdot \frac{n-2}{3} = 8\binom{n}{2} \implies \frac{n-2}{3} = 8 \implies n-2 = 24 \implies \boxed{n = 26}.$$

Question 13

[Maximum mark: 6]

No Calculator

Difficulty: Medium

Find the term independent of x in the expansion of $\frac{1}{x^3} \left(\frac{1}{3x^2} - \frac{x}{2} \right)^9$.

Question 13

Solution

The general term of $\left(\frac{1}{3x^2} - \frac{x}{2}\right)^9$ is

$$\begin{aligned}\binom{9}{r} \left(\frac{1}{3x^2}\right)^{9-r} \left(-\frac{x}{2}\right)^r &= \binom{9}{r} \left(\frac{1}{3}\right)^{9-r} \left(-\frac{1}{2}\right)^r x^{-2(9-r)+r} \\ &= \binom{9}{r} \left(\frac{1}{3}\right)^{9-r} \left(-\frac{1}{2}\right)^r x^{3r-18}.\end{aligned}$$

Multiplying by the outside factor $\frac{1}{x^3}$, the overall power of x is $3r - 18 - 3 = 3r - 21$. This is zero when $r = 7$. The term independent of x is

$$\binom{9}{7} \left(\frac{1}{3}\right)^2 \left(-\frac{1}{2}\right)^7 = 36 \times \frac{1}{9} \times \left(-\frac{1}{128}\right) = 4 \times \left(-\frac{1}{128}\right) = \boxed{-\frac{1}{32}}.$$

Question 14

[Maximum mark: 6]

No Calculator

Difficulty: Medium

The binomial expansion of $(1 + kx)^n$ is given by $1 + 12x + 28k^2x^2 + \cdots + k^n x^n$, where $n \in \mathbb{Z}$ and $k \in \mathbb{Q}$. Find the value of n and the value of k .

Question 14

Solution

By the binomial theorem,

$$(1 + kx)^n = 1 + \binom{n}{1}kx + \binom{n}{2}k^2x^2 + \dots$$

Comparing coefficients of x : $nk = 12$. (1)

Comparing coefficients of x^2 : $\binom{n}{2}k^2 = 28k^2$. Since $k \neq 0$ (otherwise the x term would vanish), divide by k^2 :

$$\binom{n}{2} = 28 \implies \frac{n(n-1)}{2} = 28 \implies n(n-1) = 56 \implies \boxed{n = 8}$$

(taking the positive integer root of $n^2 - n - 56 = 0$, i.e. $(n-8)(n+7) = 0$).

Substituting into (1): $8k = 12 \implies \boxed{k = \frac{3}{2}}$.

Question 15

[Maximum mark: 6]

Calculator

Difficulty: Medium

- (a) Express the binomial coefficient $\binom{3n+1}{3n-2}$ as a polynomial in n . [3]
- (b) Hence find the least value of n for which $\binom{3n+1}{3n-2} > 10^6$. [3]

Question 15

Solution

(a) Since $(3n + 1) - (3n - 2) = 3$, choosing $3n - 2$ items out of $3n + 1$ is the same as choosing the 3 items *not* taken:

$$\begin{aligned}\binom{3n+1}{3n-2} &= \binom{3n+1}{3} = \frac{(3n+1)(3n)(3n-1)}{3!} \\ &= \frac{3n(9n^2-1)}{6} = \boxed{\frac{9n^3-n}{2}}.\end{aligned}$$

(b) Solve $\frac{9n^3-n}{2} > 10^6$, i.e. $9n^3 - n > 2\,000\,000$. Testing values (a GDC may be used):

$$n = 60 : \frac{9(60)^3 - 60}{2} = 971\,970 (< 10^6),$$

$$n = 61 : \frac{9(61)^3 - 61}{2} = 1\,021\,384 (> 10^6).$$

So the least such value is $\boxed{n = 61}$.

Question 16

[Maximum mark: 7]

No Calculator

Difficulty: Medium

Consider the binomial expansion $(x + 1)^7 = x^7 + ax^6 + bx^5 + 35x^4 + \dots + 1$, where $x \neq 0$ and $a, b \in \mathbb{Z}^+$.

- (a) Show that $b = 21$. [2]
- (b) The third term in the expansion is the mean of the second and fourth terms in the expansion. Find the possible values of x . [5]

Question 16

Solution

(a) By the binomial theorem, $(x + 1)^7 = \sum_{r=0}^7 \binom{7}{r} x^{7-r}$. Matching powers, $a = \binom{7}{1} = 7$ (coefficient of x^6) and $b = \binom{7}{2} = 21$ (coefficient of x^5), which also matches the given coefficient $35 = \binom{7}{3}$ of x^4 as a check. Hence

$$b = \boxed{21}.$$

(b) With $a = 7$, $b = 21$, the first four terms (as numerical values once x is substituted) are

$$T_1 = x^7, \quad T_2 = 7x^6, \quad T_3 = 21x^5, \quad T_4 = 35x^4.$$

The condition "third term is the mean of the second and fourth" gives

$$21x^5 = \frac{7x^6 + 35x^4}{2} \implies 42x^5 = 7x^6 + 35x^4.$$

Since $x \neq 0$, divide by $7x^4$:

$$6x = x^2 + 5 \implies x^2 - 6x + 5 = 0 \implies (x - 1)(x - 5) = 0.$$

$$\boxed{x = 1 \text{ or } x = 5}.$$

Question 17

[Maximum mark: 7]

No Calculator

Difficulty: Hard

The coefficient of x^6 in the expansion of $(ax^3 + b)^8$ is 448. The coefficient of x^6 in the expansion of $(ax^3 + b)^{10}$ is 2880. Find the value of a and the value of b , where $a, b > 0$.

Question 17

Solution

In $(ax^3 + b)^8$, the general term is $\binom{8}{r} (ax^3)^r b^{8-r}$, giving x^6 when $3r = 6$, i.e. $r = 2$:

$$\binom{8}{2} a^2 b^6 = 448 \implies 28a^2 b^6 = 448 \implies a^2 b^6 = 16. \quad (1)$$

In $(ax^3 + b)^{10}$, similarly $r = 2$ gives x^6 :

$$\binom{10}{2} a^2 b^8 = 2880 \implies 45a^2 b^8 = 2880 \implies a^2 b^8 = 64. \quad (2)$$

Dividing (2) by (1):

$$\frac{a^2 b^8}{a^2 b^6} = \frac{64}{16} \implies b^2 = 4 \implies b = 2 \quad (b > 0).$$

Substituting into (1): $a^2(2)^6 = 16 \implies 64a^2 = 16 \implies a^2 = \frac{1}{4} \implies a = \frac{1}{2}$
($a > 0$).

$$\boxed{a = \frac{1}{2}, \quad b = 2}.$$

Question 18

[Maximum mark: 7]

No Calculator

Difficulty: Hard

The expansion of $(x + h)^8$, where $h \in \mathbb{Q}^+$, can be written as $x^8 + ax^7 + bx^6 + cx^5 + dx^4 + \cdots + h^8$, where $a, b, c, d, \dots \in \mathbb{R}$. Given that the coefficients a, b, d are the first three terms of a geometric sequence, find the value of h .

Question 18

Solution

By the binomial theorem, the coefficients of x^7, x^6, x^4 in $(x + h)^8$ are

$$a = \binom{8}{1}h = 8h, \quad b = \binom{8}{2}h^2 = 28h^2, \quad d = \binom{8}{4}h^4 = 70h^4.$$

For a, b, d to be consecutive terms of a geometric sequence, $b^2 = ad$:

$$(28h^2)^2 = (8h)(70h^4) \implies 784h^4 = 560h^5.$$

Since $h \in \mathbb{Q}^+$, $h \neq 0$, so divide both sides by h^4 :

$$784 = 560h \implies h = \frac{784}{560} = \boxed{\frac{7}{5}}.$$

Section 2

The Extended Binomial Series Negative & Fractional Indices

Questions 19–23

Every question in this section uses Formula 2 from the notes page: n is negative or a fraction, the expansion is an *infinite* series, and it is only valid for $|x| < 1$ once the bracket has been normalized to start with a bare 1 (Formula 3). Several questions here combine two expansions at once — remember the “most restrictive wins” rule (Formula 5).

Question 19

[Maximum mark: 7]

No Calculator

Difficulty: Hard

Consider the expression $\frac{1}{\sqrt{1+ax}} - \sqrt{1-x}$ where $a \in \mathbb{Q}$, $a \neq 0$. The binomial expansion of this expression, in ascending powers of x , as far as the term in x^2 , is $4bx + bx^2$, where $b \in \mathbb{Q}$.

- (a) Find the value of a and the value of b . [6]
- (b) State the restriction which must be placed on x for this expansion to be valid. [1]

Question 19

Solution

(a) Using the extended binomial series (each term already has leading coefficient 1 inside its bracket):

$$\frac{1}{\sqrt{1+ax}} = (1+ax)^{-1/2} = 1 - \frac{ax}{2} + \frac{3a^2x^2}{8} - \dots,$$

$$\sqrt{1-x} = (1-x)^{1/2} = 1 - \frac{x}{2} - \frac{x^2}{8} - \dots.$$

Subtracting,

$$\begin{aligned} \frac{1}{\sqrt{1+ax}} - \sqrt{1-x} &= \left(-\frac{a}{2} + \frac{1}{2}\right)x + \left(\frac{3a^2}{8} + \frac{1}{8}\right)x^2 + \dots \\ &= \frac{1-a}{2}x + \frac{3a^2+1}{8}x^2 + \dots \end{aligned}$$

Comparing with $4bx + bx^2$:

$$\frac{1-a}{2} = 4b, \quad \frac{3a^2+1}{8} = b.$$

Substituting the second equation into the first:

$$\frac{1-a}{2} = 4 \cdot \frac{3a^2+1}{8} = \frac{3a^2+1}{2} \implies 1-a = 3a^2+1$$

$$\implies 3a^2 + a = 0 \implies a(3a+1) = 0.$$

Since $a \neq 0$, $a = -\frac{1}{3}$. Then $b = \frac{3(1/9)+1}{8} = \frac{4/3}{8} = \frac{1}{6}$.

$$\boxed{a = -\frac{1}{3}, \quad b = \frac{1}{6}}.$$

(b) The expansion of $(1+ax)^{-1/2}$ requires $|ax| < 1$, i.e. $\left|\frac{1}{3}x\right| < 1 \iff |x| < 3$; the expansion of $(1-x)^{1/2}$ requires $|x| < 1$. Combining the two (the *most restrictive* condition sets the overall limit):

$$\boxed{|x| < 1}.$$

Question 20

[Maximum mark: 6]

No Calculator

Difficulty: Hard

In the expansion of $\frac{1}{\sqrt{q-x^2}}$, where $q \in \mathbb{Q}^+$, the coefficient of x^6 is 5120.
Find the value of q .

Question 20

Solution

First normalize so the bracket starts with a leading 1:

$$\frac{1}{\sqrt{q-x^2}} = \frac{1}{\sqrt{q}} \left(1 - \frac{x^2}{q}\right)^{-1/2}.$$

Using the extended binomial series with $n = -\frac{1}{2}$ and $u = -\frac{x^2}{q}$, the term in u^3 (which gives x^6) has coefficient

$$\binom{-1/2}{3} = \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{3!} = -\frac{5}{16}.$$

So the coefficient of x^6 in the full expression is

$$\frac{1}{\sqrt{q}} \times \left(-\frac{5}{16}\right) \times \left(-\frac{1}{q}\right)^3 = \frac{1}{\sqrt{q}} \times \left(-\frac{5}{16}\right) \times \left(-\frac{1}{q^3}\right) = \frac{5}{16q^{7/2}}.$$

Setting this equal to 5120:

$$\frac{5}{16q^{7/2}} = 5120 \implies q^{7/2} = \frac{5}{16 \times 5120} = \frac{1}{16384} = \frac{1}{2^{14}}.$$

Since $2^{14} = (2^2)^7 = 4^7$, we have $q^{7/2} = 4^{-7} = (4^{-2})^{7/2}$, so

$$\boxed{q = \frac{1}{16}}.$$

Question 21

[Maximum mark: 12]

No Calculator

Difficulty: Hard

Let $f(x) = (1 - ax)^{-1/2}$, where $ax < 1$, $a \neq 0$.

- (a) Show that the Maclaurin series for $f(x) = (1 - ax)^{-1/2}$ up to and including the x^2 term is $1 + \frac{1}{2}ax + \frac{3}{8}a^2x^2$. [2]
- (b) Hence, show that $(1 - 2x)^{-1/2}(1 - 4x)^{-1/2} \approx \frac{2 + 6x + 19x^2}{2}$. [4]
- (c) Given that the series expansion for $(1 - ax)^{-1/2}$ is convergent for $|ax| < 1$, state the restriction which must be placed on x for the approximation in (b) to be valid. [1]
- (d) Use $x = \frac{1}{10}$ to determine an approximate value for $\sqrt{3}$. Give your answer in the form $\frac{c}{d}$, where $c, d \in \mathbb{Z}^+$. [5]

Question 21

Solution

(a) Applying the extended binomial series $(1 + u)^n = 1 + nu + \frac{n(n-1)}{2!}u^2 + \dots$ with $n = -\frac{1}{2}$ and $u = -ax$:

$$\begin{aligned}(1 - ax)^{-1/2} &= 1 + \left(-\frac{1}{2}\right)(-ax) + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2}(-ax)^2 + \dots \\ &= 1 + \frac{ax}{2} + \frac{3}{8}a^2x^2 + \dots\end{aligned}$$

as required, so $f(x) \approx \boxed{1 + \frac{1}{2}ax + \frac{3}{8}a^2x^2}$.

(b) Applying the result of (a) with $a = 2$ (for the factor $(1 - 2x)^{-1/2}$) and $a = 4$ (for $(1 - 4x)^{-1/2}$):

$$\begin{aligned}(1 - 2x)^{-1/2} &\approx 1 + x + \frac{3}{8}(4)x^2 = 1 + x + \frac{3}{2}x^2, \\ (1 - 4x)^{-1/2} &\approx 1 + 2x + \frac{3}{8}(16)x^2 = 1 + 2x + 6x^2.\end{aligned}$$

Multiplying and keeping terms up to x^2 :

$$\begin{aligned}(1 + x + \frac{3}{2}x^2)(1 + 2x + 6x^2) &= 1 + 2x + 6x^2 + x + 2x^2 + \frac{3}{2}x^2 + O(x^3) \\ &= 1 + 3x + \frac{19}{2}x^2 + \dots\end{aligned}$$

Multiplying numerator and denominator by 2:

$$(1 - 2x)^{-1/2}(1 - 4x)^{-1/2} \approx \boxed{\frac{2 + 6x + 19x^2}{2}}.$$

(c) The factor $(1 - 2x)^{-1/2}$ needs $|2x| < 1 \iff |x| < \frac{1}{2}$; the factor $(1 - 4x)^{-1/2}$ needs $|4x| < 1 \iff |x| < \frac{1}{4}$. Since *both* expansions must be valid simultaneously, the **most restrictive** of the two conditions sets the overall limit:

$$\boxed{|x| < \frac{1}{4}}.$$

(Notice $x = \frac{1}{10}$ used in part (d) safely satisfies this.)

(d) At $x = \frac{1}{10}$ (which satisfies $|x| < \frac{1}{4}$ from (c)), the exact value is

$$\begin{aligned}(1 - 0.2)^{-1/2}(1 - 0.4)^{-1/2} &= \frac{1}{\sqrt{0.8 \times 0.6}} = \frac{1}{\sqrt{0.48}} = \frac{1}{\sqrt{12/25}} \\ &= \frac{5}{2\sqrt{3}} = \frac{5\sqrt{3}}{6}.\end{aligned}$$

Using the approximation from (b):

$$\frac{2 + 6(0.1) + 19(0.01)}{2} = \frac{2 + 0.6 + 0.19}{2} = \frac{2.79}{2} = \frac{279}{200}.$$

Setting these equal to solve for $\sqrt{3}$:

$$\frac{5\sqrt{3}}{6} \approx \frac{279}{200} \implies \sqrt{3} \approx \frac{279}{200} \times \frac{6}{5} = \frac{1674}{1000} = \boxed{\frac{837}{500}}.$$

Question 22

[Maximum mark: 17]

No Calculator

Difficulty: Hard

- (a) Find the first four terms in the binomial expansion of $\sqrt{1+5x}$ in ascending powers of x . [4]

Consider the expression $(1+px)(1+qx)^{-1}$, where $p, q \in \mathbb{Q}$.

- (b) Find the expansion of $(1+px)(1+qx)^{-1}$ in ascending powers of x , up to and including the term in x^2 . [3]

The expansions found in parts (a) and (b) are identical up to the first three terms, for a value of p and a value of q .

- (c) Show that $q = \frac{5}{4}$. [4]

The expression $\frac{1+px}{1+qx}$, with $p = \frac{15}{4}$ and $q = \frac{5}{4}$, can be used as an approximation for $\sqrt{1+5x}$ where $|x| < \frac{1}{5}$.

- (d) (i) Hence, by finding a suitable value for x , find the approximation for $\sqrt{1.2}$ in the form $\frac{m}{n}$, where $m, n \in \mathbb{Z}$.

- (ii) Now consider the approximation for $\frac{\sqrt{5}}{2}$. Explain why the approximation for $\frac{\sqrt{5}}{2}$ is not as accurate as the approximation for $\sqrt{1.2}$.

[6]

Question 22

Solution

(a) Using the extended binomial series with $n = \frac{1}{2}$, $u = 5x$:

$$\begin{aligned}(1 + 5x)^{1/2} &= 1 + \frac{1}{2}(5x) + \frac{\frac{1}{2}\left(-\frac{1}{2}\right)}{2!}(5x)^2 + \frac{\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!}(5x)^3 + \dots \\ &= 1 + \frac{5}{2}x - \frac{1}{8}(25x^2) + \frac{1}{16}(125x^3) + \dots = \boxed{1 + \frac{5}{2}x - \frac{25}{8}x^2 + \frac{125}{16}x^3}.\end{aligned}$$

(b) $(1 + qx)^{-1} = 1 - qx + q^2x^2 - \dots$, so

$$\begin{aligned}(1 + px)(1 + qx)^{-1} &= (1 + px)(1 - qx + q^2x^2 - \dots) \\ &= 1 + (p - q)x + (q^2 - pq)x^2 + \dots \\ &= \boxed{1 + (p - q)x + q(q - p)x^2}.\end{aligned}$$

(c) Matching the x -coefficients of (a) and (b): $p - q = \frac{5}{2}$. Matching the x^2 -coefficients: $q(q - p) = -\frac{25}{8}$, i.e. $q\left(-\frac{5}{2}\right) = -\frac{25}{8}$, so

$$q = \frac{25/8}{5/2} = \frac{25}{8} \times \frac{2}{5} = \boxed{\frac{5}{4}}.$$

(Then $p = q + \frac{5}{2} = \frac{5}{4} + \frac{10}{4} = \frac{15}{4}$, consistent with the value given for part (d).)

(d)(i) We need $1 + 5x = 1.2 \implies x = 0.04 = \frac{1}{25}$, which satisfies $|x| < \frac{1}{5}$.

Then

$$\left. \frac{1 + \frac{15}{4}x}{1 + \frac{5}{4}x} \right|_{x=1/25} = \frac{1 + \frac{15}{100}}{1 + \frac{5}{100}} = \frac{1.15}{1.05} = \frac{115}{105} = \boxed{\frac{23}{21}}.$$

(d)(ii) Note $\frac{\sqrt{5}}{2} = \sqrt{\frac{5}{4}} = \sqrt{1 + 5x}$ with $1 + 5x = 1.25 \implies x = 0.05 = \frac{1}{20}$, which also satisfies $|x| < \frac{1}{5}$, so the approximation is still valid — but $|x| = \frac{1}{20} = 0.05$ is **larger** than the value $|x| = \frac{1}{25} = 0.04$ used for $\sqrt{1.2}$. Since the truncation error of a series expansion grows as $|x|$ moves further

from 0, using a larger $|x|$ produces a less accurate approximation. Hence the approximation for $\frac{\sqrt{5}}{2}$ is

less accurate — larger $|x|$

(though still valid) gives a larger truncation error.

Question 23

[Maximum mark: 18]

No Calculator

Difficulty: Hard

Consider the expression $\frac{1}{1+x^2}$, where $|x| < 1$.

- (a) (i) Show that the first four terms in the binomial expansion of $\frac{1}{1+x^2}$ are $1 - x^2 + x^4 - x^6$. [3]
- (ii) Hence, find an approximation for $\int \frac{1}{1+x^2} dx$ up to and including the term in x^7 . [2]

Now consider the expression $\frac{1}{\sqrt{1-x^2}}$, where $|x| < 1$.

- (b) (i) Expand $\frac{1}{\sqrt{1-x^2}}$ up to and including the term in x^4 .
- (ii) Hence, find a polynomial expression for $\int \frac{1}{\sqrt{1-x^2}} dx$ up to and including the term in x^5 . [5]

The expression $\frac{25}{48} + \frac{k}{1280}$, where $k \in \mathbb{Z}^+$, can be used to approximate $\arcsin \frac{1}{2}$.

- (c) Use the result from part (b)(ii) to find the value of k . [4]
- (d) Hence, find an approximation for π in the form $\frac{p}{640}$ where $p \in \mathbb{Z}^+$. [4]

Question 23

Solution

(a)(i) Using the extended series with $n = -1$, $u = x^2$:

$$\frac{1}{1+x^2} = (1+x^2)^{-1} = 1 - x^2 + x^4 - x^6 + \dots \quad \text{as required.}$$

(a)(ii) Integrating term by term:

$$\int \frac{1}{1+x^2} dx \approx \boxed{x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} (+C)}.$$

(b)(i) Using $n = -\frac{1}{2}$, $u = -x^2$:

$$\begin{aligned} (1-x^2)^{-1/2} &= 1 + \left(-\frac{1}{2}\right)(-x^2) + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2}(-x^2)^2 + \dots \\ &= \boxed{1 + \frac{x^2}{2} + \frac{3x^4}{8}}. \end{aligned}$$

(b)(ii) Integrating term by term:

$$\int \frac{1}{\sqrt{1-x^2}} dx \approx \boxed{x + \frac{x^3}{6} + \frac{3x^5}{40} (+C)}.$$

(c) Since $\arcsin x = \int \frac{1}{\sqrt{1-x^2}} dx$ (with constant of integration 0, as $\arcsin 0 = 0$), the result of (b)(ii) gives

$$\arcsin \frac{1}{2} \approx \frac{1}{2} + \frac{(1/2)^3}{6} + \frac{3(1/2)^5}{40} = \frac{1}{2} + \frac{1}{48} + \frac{3}{1280}.$$

Combine over a common denominator of 3840: $\frac{1}{2} = \frac{1920}{3840}$, $\frac{1}{48} = \frac{80}{3840}$,

$\frac{3}{1280} = \frac{9}{3840}$, giving

$$\arcsin \frac{1}{2} \approx \frac{1920 + 80 + 9}{3840} = \frac{2009}{3840}.$$

Matching with $\frac{25}{48} + \frac{k}{1280}$: note $\frac{25}{48} = \frac{2000}{3840}$, so

$$\frac{2009}{3840} = \frac{2000}{3840} + \frac{k}{1280} \implies \frac{9}{3840} = \frac{k}{1280} \implies k = \frac{9 \times 1280}{3840} = \boxed{3}.$$

(d) Since $\arcsin \frac{1}{2} = \frac{\pi}{6}$, and using the approximation $\arcsin \frac{1}{2} \approx \frac{25}{48} + \frac{3}{1280} = \frac{2009}{3840}$ from part (c):

$$\pi \approx 6 \times \frac{2009}{3840} = \frac{2009}{640}.$$

$$\boxed{p = 2009}.$$